



Probabilistic Multi-agent Only-believing

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ABSTRACT

Levesque introduced the notion of only-knowing to precisely capture the beliefs of a knowledge base. While numerous studies of only-knowing have emerged, such as the representation of probabilistic beliefs or reasoning about beliefs in an uncertain dynamical system, most remain confined to single-agent contexts. This limitation predominantly stems from an absence of a logical framework, which faithfully extends Levesque’s intuition of only-knowing to multi-agent, probabilistic scenarios.

In this paper, we introduce a first-order logical account with probabilistic beliefs and only-believing of many agents. We demonstrate that the categorical fragment of our account forms a $KD45_n$ modal system, and the notion of belief has behavior following the laws of probability. We also show how an agent’s beliefs and non-beliefs about the environment or other agents’ beliefs are precisely captured through the modalities of only-believing, which paves the way to generalize tools for interfacing with symbolic, probabilistic knowledge bases. By way of example, we demonstrate how non-monotonic conclusions including default reasoning can be handled by our account.

KEYWORDS

knowledge representation; modal epistemic logic; only-believing; multi-agent systems; reasoning about probability

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1 INTRODUCTION

In multi-agent systems, where agents constantly interact or coordinate, reasoning about knowledge and belief is of interest for many applications. By gathering detailed information, not only from the environment but also from other agents’ mental states, an agent can determine when communication with other agents is necessary, enhance collaboration on tasks, and optimize its strategy against other agents in scenarios where agents operate concurrently.

Among many symbolic approaches that represent an agent’s knowledge and belief, the notion of only-knowing [23, 24] is particularly valuable: An agent’s beliefs of a knowledge base (KB) are modelled in terms of only-knowing a collection of sentences, and

sentences that are not logically entailed by the KB are taken to be precisely those not believed. In Levesque’s logic of only-knowing OL [23, 24], the classical epistemic operator K is used for knowledge, and in addition, a modality O is used for only-knowing. To illustrate, Op means that p is all the agent knows. Op entails Kp , but it also entails $\neg Kq$ and $\neg K\neg q$ for proposition q different from p . This is different from classical epistemic logic where Kp entails neither $\neg Kq$ nor $\neg K\neg q$. Furthermore, only-knowing also shows a close connection to autoepistemic logic [27] and can be applied to autoepistemic reasoning or default reasoning [8]. With some simple augmentations, non-monotonic conclusions can be reached without using meta-logical notions such as fixpoints or partial orders [20, 31]. For instance, let KB be a single sentence $\{Bird(tweety)\}$, namely “tweety is a bird”, δ a sentence to express the default $\forall x. [Bird(x) \wedge \neg K\neg Fly(x) \supset Fly(x)]$, i.e. any bird is assumed to be able to fly unless we know the opposite. Then we have the following non-monotonic properties in OL : $O(KB \wedge \delta)$ entails $KFly(tweety)$, but $O(KB \wedge \delta \wedge \neg Fly(tweety))$ entails $K\neg Fly(tweety)$. In other words, the initial belief “tweety can fly” is retracted when a new fact is added.

Numerous researches on single-agent only-knowing emerged: Lakemeyer and Levesque extended the notion of only-knowing to capture different forms of default reasoning [20], Belle et al. proposed the logic $OB\mathcal{L}$ to describe *only-believing* and admit knowledge bases with incomplete, probabilistic specifications [6]. There is also work on reasoning in dynamical domains [5, 19, 21]. It has also been shown how to capture an agent’s belief after actions via only-believing and how to perform projection reasoning [25, 26].

Naturally, extending these works into the multi-agent scenario would result in an expressive account to represent and reason about the mental states of agents, and about how their minds change as a result of actions. As for non-monotonic reasoning, via multi-agent only-believing one could expect to express default assumption not only on the facts (e.g. the aforementioned “Birds can fly” example) but also on other agents’ beliefs, like “It is assumed that agent 2 believes that the coin is fair”. The expressiveness of such an account should be of interest for planning and decision-making in cooperative or concurrent games. However, research into multi-agent only-knowing has faced more obstacles than expected, and extensions in terms of probabilistic belief or belief after actions are rarely considered yet. Most studies are confined to characterizing only-knowing of categorical knowledge. Halpern and Lakemeyer attempted to handle the extension independently [13, 18]. However, these accounts make use of arbitrary Kripke structures and lose the simplicity of Levesque’s semantics. Furthermore, each account has some undesirable properties respectively. These properties are avoided in a joint work from Halpern and Lakemeyer [14], but it forces us to have the semantic notion of validity directly in the language. For this reason, that proposal is not natural, and it is



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matched with a proof theory that has a set of new axioms to deal with these new notions. Belle and Lakemeyer revisited the construction and proposed an account $ON\mathcal{L}_n$ with a natural possible-world semantics[3]. However, their notion of only-knowing didn't precisely capture the belief and non-belief for sentences of higher depth of nesting. For example, let O_1 denotes the only-knowing of agent 1, K_1 and K_2 stand for the belief of agent 1 and 2 respectively, then $O_1p \supset K_1(K_2q)$ and $O_1p \supset \neg K_1(K_2q)$ are both satisfiable in their semantics.

The major difficulty is as follows: If we faithfully follow Levesque's principle of only-knowing, that the beliefs of an agent are precisely those following from its knowledge base, then in the multi-agent cases, only-knowing a sentence involves the belief and non-belief of sentences with an arbitrary depth of nesting beliefs. In a single-agent scenario, e.g. OL , every formula specifies the agent's belief about the fact, or about its own belief, which we say to be of depth 1. In a multi-agent scenario, however, there are sentences describing an agent's belief about another agent's belief, which are of depth 2 at least. A sentence describing agent 1's belief about agent 2's belief about agent 3's belief is at least of depth 3. Such a belief nesting can be arbitrarily deep. If the depth is not restricted, constructing a structure which describes only-knowing up to all depths will be semantically difficult and will easily lead to a circular definition.

The study of belief with arbitrary depths of nesting is indeed meaningful, in particular for reasoning about common knowledge. However, we argue that it suffices to consider finite depths in many applications, where common knowledge or beliefs on an infinite set of sentences are not considered. We adapt the idea from Aucher and Belle [1] and use modalities of the form $O_i^{(k)}$ to describe agent i 's only-believing up to depth k . We focus on extending the notion of only-knowing to multi-agent, probabilistic cases and spare us from the troubles handling infinite nesting of beliefs.

The rest of the paper is organized as follows. We begin with introducing a new logic of multi-agent only-believing $OB\mathcal{L}_m$. Then we analyze the properties of the logic, in particular those exclusive for only-believing. Relation to other logical accounts will also be discussed. Finally, we briefly conclude our work.

2 THE LOGIC $OB\mathcal{L}_m$

We introduce the logic $OB\mathcal{L}_m$. It can be considered as a multi-agent extension of $OB\mathcal{L}$ [6] and a probabilistic extension of $ON\mathcal{L}_n$ [3].

2.1 The Language

The logic $OB\mathcal{L}_m$ is a first-order modal logic with equality. Let $Ag = \{1, \dots, m\}$ denote a set of m agents. The vocabulary consists of (FO) *variables* and *predicate symbols*. For simplicity, function symbols are excluded. The language includes a countable set of *standard names* $\mathcal{N}$, which are syntactically treated like constants. This can be viewed as having fixed infinite domain closure axioms with the *unique name assumption*, which further allows FO quantification to be understood substitutionally. The set of rational numbers $\mathbb{Q}$ is included as a sub-sort of standard names. We call a predicate other than $=$, applied to first-order variables or standard names, an *atomic formula*. An atomic formula without variables is called a *ground atom*.

Standard FO connectives $\wedge, \neg, \forall$ and modalities $B_1, \dots, B_m$ are used to construct formulae. For each agent index i , $B_i(\alpha: r)$ is read as "agent i believes α with degree r ". To illustrate, a sentence $B_1(Rain: 0.8)$ means agent 1 believes that there is an 80% chance of raining. In particular, we write $K_i\alpha$ as an abbreviation of $B_i(\alpha: 1)$ and read it as "agent i knows α ".

To extend the notion of "only-knowing" to multi-agent systems, we start from the single-agent cases and inductively extend the language to represent only-believing of any depth. We use modality $O_i^{(1)}$ to specify agent i 's only-believing up to depth 1, which captures the beliefs and non-beliefs about facts and the agent's own beliefs. An example of $O_i^{(1)}$ -formula is $O_i^{(1)}(\exists x.[P(x) \wedge \neg K_i P(x)])$, which means "All agent i believes up to depth 1 is the existence of an unknown object x s.t. $P(x)$ holds." Now we go one step deeper: We use modality $O_i^{(2)}$ to specify agent i 's only-believing up to depth 2, which stands for the beliefs and non-beliefs about fact, about the agent's own beliefs and other agents' beliefs up to depth 1. An example is

$$O_i^{(2)}(q \wedge \exists x.[K_j(P(x)) \wedge \neg K_i K_j(P(x))])$$

with $i \neq j$. The difference between $O_i^{(2)}$ and $O_i^{(1)}$ is: While the former specifies agent i 's beliefs about other agents' beliefs, the latter does not. For instance, $O_i^{(2)}(p: 1)$ implies that agent i knows nothing about what other agents know or believe, e.g. $O_i^{(2)}(p: 1)$ entails $\neg K_i K_j q$ for any proposition q , but $O_i^{(1)}(p: 1)$ entails neither $K_i K_j q$ nor $\neg K_i K_j q$.

We include modalities $O_1^{(k)}, \dots, O_m^{(k)}$ with natural number $k > 0$. For each k , $O_i^{(k+1)}$ specifies agent i 's only-believing about other agents' beliefs up to depth k , and formula $O_i^{(k)}(\alpha_1: r_1, \dots, \alpha_l: r_l)$ is read as "All agent i believes up to depth k is: α_1 with degree r_1 , ..., and α_l with degree r_l ". We write $O_i^{(k)}\alpha$ to mean $O_i^{(k)}(\alpha: 1)$.

DEFINITION 1 (*i*-DEPTH). For $i \in Ag$, the *i*-depth of a formula α , written as $dep[\alpha, i]$, is inductively defined as

- $dep[\alpha, i] = 0$ for atomic formula α , $dep[t_1 = t_2, i] = 0$
- $dep[\neg\alpha, i] = dep[\alpha, i]$
- $dep[\alpha \wedge \beta, i] = \max(dep[\alpha, i], dep[\beta, i])$
- $dep[\forall x.\alpha, i] = dep[\alpha, i]$
- $dep[B_i(\alpha: r), i] = \max\{\max\{dep[\alpha, j] \mid j \neq i\} + 1, dep[\alpha, i]\}$
- $dep[B_j(\alpha: r), i] = 0$ for $j \neq i$
- $dep[O_i^{(k)}(\alpha_1: r_1, \dots, \alpha_l: r_l), i] = k$
- $dep[O_j^{(k)}(\alpha_1: r_1, \dots, \alpha_l: r_l), i] = 0$ for $j \neq i$

For a B -formula, the depth increases only when a nesting of different agents' beliefs occurs. For an O -formula, the depth is given by the superscript k .

EXAMPLE 1. Let α denote the formula $(\exists r. r > 0.5 \wedge B_2(p: r))$. Consider the formula $K_1\alpha$, namely, agent 1 knows that p is believed by agent 2 with a degree greater than 0.5. Then

$$\begin{aligned} dep[K_1\alpha, 1] &= \max\{dep[\alpha, 1], 1 + dep[\alpha, 2]\} \\ &= \max\{0, 2\} = 2 \end{aligned}$$

$$dep[K_1\alpha, 2] = 0$$

i.e. $K_1\alpha$ has 1-depth 2 and 2-depth 0.

The set of well-formed formulae (wffs) is the smallest set including:

- any atomic formulae,
- $t = t'$ where t and t' are variables or standard names,
- if α, β are formulae, then $\alpha \wedge \beta$, $\neg\alpha$, $\forall x.\alpha$ are formulae
- if α is a formula, then for any $i \in \{1, \dots, m\}$ and number r , $B_i(\alpha : r)$ is a formula.
- for $j \in \{1, \dots, l\}$, $B_i(\alpha_j : r_j)$ is a formula with i -depth k_j , $k \geq \max\{k_1, \dots, k_l\}$, then $O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ is a formula.

An example of illegal formulae is $O_1^{(1)}(p \wedge K_2 q : 0.8)$, where the sentence being believed is of 1-depth 2. Only-believing this sentence requires at least $k = 2$. For the rest of the paper, we only consider wffs and proper superscript k .

A formula with no free variables is called a *sentence*. A formula not mentioning modalities is said to be *objective*. A formula, where all predicate symbols appear within the scope of a modal operator is called *subjective*. Given $i \in \mathcal{A}$, a formula is called *i-objective* if modal operators B_i and $O_i^{(k)}$ only appear within the scope of B_j or $O_j^{(k')}$ where $j \neq i$. For instance, $p \wedge O_2^{(3)}(q \wedge K_1 q)$ is 1-objective since K_1 only appears in the scope of $O_2^{(3)}$, namely, knowing p is agent 2's conjecture of agent 1's mental state and not necessarily to be the actual belief of agent 1. Inversely, a formula is called *i-subjective* if any predicates or modal operators B_j and $O_j^{(k)}$ s.t. $j \neq i$ only appear within the scope of operators B_i or $O_i^{(k')}$, e.g. $\neg K_1 p \wedge B_1(p \wedge K_2 q : 0.5)$. We use True to denote an objective tautology $\forall x.(x = x)$ and use False to denote its negation.

2.2 The Semantics

The semantics of $OB\mathcal{L}_m$ is given in terms of possible worlds, where a *world* is a set of ground atoms considered as true. The set of all worlds is denoted as $\mathcal{W}$.

In the logic $OB\mathcal{L}$, a distribution d is a function from $\mathcal{W}$ to the set of non-negative real numbers $\mathbb{R}^{\geq 0}$ and an epistemic state is defined as a set of distributions.¹ We extend the notions into multi-agent cases: A distribution d defined above describes an agent's weighting to each possible world. For an agent i , to describe agent i 's belief about other agents' beliefs on the actual world, the weight assigned to world w , namely $d(w)$, is partitioned and reassigned to tuples of form $(w, e_1, \dots, e_{m-1})$ where each e_j is an $OB\mathcal{L}$ epistemic state describing the belief of an agent other than i . Formally, we have the following definition:

DEFINITION 2 (k -DISTRIBUTION). For $k \geq 1$, a k -distribution, written as d^k , is inductively defined as:

- $d^1 : (\mathcal{W} \times \underbrace{\{\emptyset\} \times \dots \times \{\emptyset\}}_{m-1}) \rightarrow \mathbb{R}^{\geq 0}$,
- $d^{k+1} : (\mathcal{W} \times \underbrace{\mathcal{E}^k \times \dots \times \mathcal{E}^k}_{m-1}) \rightarrow \mathbb{R}^{\geq 0}$,

where $\mathcal{E}^k = 2^{\mathcal{D}^k}$, and $\mathcal{D}^k$ is the set of all k -distributions.

¹The idea to incorporate a set of distributions instead of a single distribution in the epistemic state derives from the philosophical stance that *de re* knowledge about degrees of belief should not be valid. Namely, if epistemic states only contain single distributions, formulae such as $\exists r.K(B(\phi : r))$ will be valid for any ϕ , which is counter-intuitive.

When the context is clear, we omit the superscript and write d instead of d^k . We say e is an *epistemic state*, specifically a k -epistemic state, if e is a set of k -distributions, i.e. $e \in \mathcal{E}^k$. The number k is said to be the depth of e , written as $\text{dep}[e] = k$.

Similar to $OB\mathcal{L}$, we appeal to three conditions BND, EQ, NORM to obtain probability distributions.

DEFINITION 3. Let d be a $(k+1)$ -distribution for some $k \geq 0$, $\mathcal{U}, \mathcal{V} \subseteq (\mathcal{W} \times \mathcal{E}^k \times \dots \times \mathcal{E}^k)$ and $\mathcal{U} \subseteq \mathcal{V}$, r any real number. We define conditions BND, EQ and NORM as follows:

- $\text{BND}(d, \mathcal{U}, r)$ iff there is no l , $\{(w_l, \vec{e}_1), \dots, (w_l, \vec{e}_l)\} \subseteq \mathcal{U}$ s.t.²

$$\sum_{j=1}^l d(w_j, \vec{e}_j) > r$$

- $\text{EQ}(d, \mathcal{U}, r)$ iff $\text{BND}(d, \mathcal{U}, r)$ and no $r' < r$ s.t. $\text{BND}(d, \mathcal{U}, r')$;
- $\text{NORM}(d, \mathcal{U}, \mathcal{V}, r)$ iff there is a number $b \neq 0$ such that $\text{EQ}(d, \mathcal{U}, b \times r)$ and $\text{EQ}(d, \mathcal{V}, b)$.

Intuitively, $\text{BND}(d, \mathcal{U}, r)$ ensures the weight of tuples $(w, \vec{e})$ in $\mathcal{U}$ wrt d is bounded by r . $\text{EQ}(d, \mathcal{U}, r)$ ensures that the weight is bounded and r is the supremum. Given $\text{NORM}(d, \mathcal{U}, \mathcal{V}, r)$, r can be viewed as the normalized sum of the weight of worlds in $\mathcal{U}$ in relation to $\mathcal{V}$. Essentially, although the distribution d is defined over an uncountable domain, these conditions on d admit a well-defined summation, and the weights on worlds can indeed be interpreted as probabilities:

THEOREM 1. For $k \geq 0$, suppose that d is a $(k+1)$ -distribution. Let $\mathcal{V} = (\mathcal{W} \times \mathcal{E}^k \times \dots \times \mathcal{E}^k)$ and $\mathcal{U} = \{(w, \vec{e}) \mid d(w, \vec{e}) \neq 0\}$. For any $b \geq 0$, if $\text{BND}(d, \mathcal{V}, b)$, then $\mathcal{U}$ is countable.

PROOF. Let $\mathcal{U}_j = \{(w, \vec{e}) \in \mathcal{U} \mid d(w, \vec{e}) \geq 1/j\}$ for $j \in \mathbb{N}^+$. It is easy to see that $\mathcal{U} = \bigcup \mathcal{U}_j$. Suppose that $\mathcal{U}$ is uncountable, then there is some $\epsilon > 0$ such that $\mathcal{U}_\epsilon = \{w \in \mathcal{U} \mid d(w, \vec{e}) \geq \epsilon\}$ is infinite (Otherwise we could enumerate $\mathcal{U}$ by enumerating $\mathcal{U}_j$ starting at $j = 1$). Consider any countably infinite sequence $(w_l, \vec{e}_l)$ taken from $\mathcal{U}_\epsilon$. Since $d(w_l, \vec{e}_l) \geq \epsilon$ for all l , the sum $\sum_{l=1}^\infty d(w_l, \vec{e}_l)$ is unbounded, contradicting the assumption that $\mathcal{V}$ is bounded. $\square$

By satisfying the BND conditions, a distribution assigns non-zero values to a countable support set $\text{Supp}(d) = \{(w, \vec{e}) \mid d(w, \vec{e}) \neq 0\}$. For the rest of the paper, by summation of weights of a bounded distribution d , we always mean summation in its support set, i.e.

$$\sum_{(w, \vec{e}) \in E} d(w, \vec{e}) \doteq \sum_{(w, \vec{e}) \in E \cap \text{Supp}(d)} d(w, \vec{e})$$

In $OB\mathcal{L}$, the necessitation rule does not hold for modality B (also K). For instance, True is valid in $OB\mathcal{L}$ yet $K\text{True}$ is not. Besides, an empty epistemic state is legal in $OB\mathcal{L}$, which breaks the laws of probability (e.g. $B(p : 0.1) \wedge B(p : 0.2)$ will be satisfied by the empty state). Fortunately, these undesirable properties can be easily avoided by ruling out improper distributions and epistemic states.

DEFINITION 4 (REGULARITY). We inductively define the sets of **regular k -distributions** $\mathcal{D}^k$ and **regular k -epistemic states** $\mathcal{E}^k$: Let $\mathcal{E}^0 = \{\emptyset\}$. For any $k > 0$, let $\mathcal{V} = (\mathcal{W} \times \mathcal{E}^{k-1} \times \dots \times \mathcal{E}^{k-1})$,

²We write the arguments of a k -distribution as $(w, \vec{e})$ instead of $(w, e_1, \dots, e_{m-1})$.

- $\mathcal{D}^k = \{d \mid EQ(d, \mathcal{V}, 1)\}$;
- $\mathcal{E}^k = 2^{\mathcal{D}^k} \setminus \{\emptyset\}$.

For the rest of the paper, we consider only regular distributions and epistemic states. Now we define the truth of sentences. By a model, we mean a tuple $(w, \vec{e})$, where $\vec{e} = (e_1, \dots, e_m)$ and each e_i is an epistemic state (e_i denotes the i -th argument of $\vec{e}$). We say a formula α and $\vec{e}$ are *compatible* if $\text{dep}[\alpha, i] \leq \text{dep}[e_i]$ f.a. $i \in \{1, \dots, m\}$. The truth value of objective sentences is assigned as follows:

- $w, \vec{e} \models P(n)$ iff $P(n) \in w$;
- $w, \vec{e} \models t_1 = t_2$ iff t_1 and t_2 are identical standard name;
- $w, \vec{e} \models \neg\alpha$ iff not $w, \vec{e} \models \alpha$;
- $w, \vec{e} \models \alpha \wedge \beta$ iff $w, \vec{e} \models \alpha$ and $w, \vec{e} \models \beta$;
- $w, \vec{e} \models \forall x.\alpha$ iff $w, \vec{e} \models \alpha_n^x$ for any $n \in \mathcal{N}$.

Here α_n^x means the formula obtained by substituting each appearance of free variable x in α by a standard name n . The semantics for the objective fragment is identical to language $\mathcal{L}$ [24]. Supposing that $\vec{e}$ is compatible with $B_i(\alpha : r)$, then

- $w, \vec{e} \models B_i(\alpha : r)$ iff f.a. $d \in e_i$, $\text{NORM}(d, \mathcal{W}_\alpha^{i, e_i}, \mathcal{W}_{\text{TRUE}}^{i, e_i}, r)$,

where for $i \in \text{Ag}$, $e \in \mathcal{E}^k$ and formula φ , $\mathcal{W}_\varphi^{i, e}$ is defined as

$$\mathcal{W}_\varphi^{i, e} = \{(w, \vec{e}_{-i}) \mid w, e_1, \dots, e_{i-1}, e, e_{i+1}, \dots, e_m \models \varphi, \\ e_j \in \mathcal{E}^{k-1} \text{ for } j \neq i\}$$

and $\vec{e}_{-i} = (e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_m)$. Apparently, for any i and $e \in \mathcal{E}^{k+1}$, $\mathcal{W}_{\text{TRUE}}^{i, e} = (\mathcal{W} \times \mathcal{E}^k \times \dots \times \mathcal{E}^k)$. We drop the superscript and write $\mathcal{W}_{\text{TRUE}}$ for simplicity.

Given an epistemic state of depth k , we can define the semantics of only-believing up to depth k :

- For $e_i \in \mathcal{E}^k$, $w, \vec{e} \models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ iff f.a. $d \in \mathcal{D}^k$, $d \in e_i$ iff $\text{NORM}(d, \mathcal{W}_{\alpha_j}^{i, e_i}, \mathcal{W}_{\text{TRUE}}^{i, e_i}, r_j)$ for $j \in \{1, \dots, l\}$.

We still need to assign truth values for only-believing up to depth k , while the model is of a greater depth (otherwise sentences like $O_1^{(1)} p \wedge O_1^{(2)} (p \wedge K_2 p)$ are unsatisfiable). To do so, we need to capture a model's belief of lower depths and omit the depths greater than needed. This is achieved via the notion which we call "regression":

DEFINITION 5 (REGRESSION). We inductively define the regression of any distributions and epistemic states:

- For $d \in \mathcal{D}^2$, $d' \in \mathcal{D}^1$, d' is the regression of d , written as $d' = \mathbf{R}[d]$ if for any w ,
- $$\sum_{e_1^*, \dots, e_{m-1}^*} d(w, e_1^*, \dots, e_{m-1}^*) = d'(w, \emptyset, \dots, \emptyset)$$
- For $e \in \mathcal{E}^2$, $e' \in \mathcal{E}^1$, we say e' is the regression of e , written as $e' = \mathbf{R}[e]$, iff $e' = \{\mathbf{R}[d] \mid d \in e\}$
 - For $d \in \mathcal{D}^{k+1}$, $d' \in \mathcal{D}^k$, d' is the regression of d , written as $d' = \mathbf{R}[d]$, if for any $w, e_1', \dots, e_{m-1}'$

$$\sum_{\tilde{e}_1, \dots, \tilde{e}_{m-1} \in E} d(w, \tilde{e}_1, \dots, \tilde{e}_{m-1}) = d'(w, e_1', \dots, e_{m-1}')$$

where $E = \{\tilde{e}_1, \dots, \tilde{e}_{m-1} \mid e_1' = \mathbf{R}[\tilde{e}_1], \dots, e_{m-1}' = \mathbf{R}[\tilde{e}_{m-1}]\}$

- For $e \in \mathcal{E}^{k+1}$, $e' \in \mathcal{E}^k$, we say e' is the regression of e , written as $e' = \mathbf{R}[e]$, iff $e' = \{\mathbf{R}[d] \mid d \in e\}$

By definition, for each $e \in \mathcal{E}^{k+1}$, there exists a unique $e' \in \mathcal{E}^k$ s.t. $e' = \mathbf{R}[e]$. We write $e' = \mathbf{R}^{(2)}[e]$ to mean $e' = \mathbf{R}[\mathbf{R}[e]]$. Analogously, $e' = \mathbf{R}^{(k)}[e]$ means $e' = \mathbf{R}[\dots \mathbf{R}[e] \dots]$ with k nested $\mathbf{R}$. The following lemma indicates that $\mathbf{R}[e]$ faithfully reflects the properties of e in lower depths. Therefore, for $e \in \mathcal{E}^{k+1}$, it is reasonable to assign truth values for only-believing up to depth k based on the truth assignment of $\mathbf{R}[e]$.

LEMMA 1. Given $\vec{e}$ and $\vec{e}'$ s.t. $e_i' = \mathbf{R}[e_i]$ f.a. $i \in \{1, \dots, m\}$. For any formula α compatible with $\vec{e}'$ and mentioning no O -operators, $w, \vec{e} \models \alpha$ iff $w, \vec{e}' \models \alpha$.

PROOF. We prove the lemma via induction.

Basis:

For objective α , the proof is trivial since the truth value is irrelevant to $\vec{e}$ and $\vec{e}'$.

Induction hypothesis:

Suppose that the statement holds for any $\vec{e}, \vec{e}', \alpha'$ s.t. $e_i' = \mathbf{R}[e_i]$, $e_i' \in \mathcal{E}^l$ for some $l < k_i$ and α' compatible with $\vec{e}'$.

- Induction on $\wedge, \neg$ or $\forall$ is trivial.
- For $e_i \in \mathcal{E}^{k_i+1}$, $e_i \models B_i(\alpha : r)$ iff f.a. $d \in e_i$,

$$\sum_{(w, \vec{e}_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w, \vec{e}_{-i}) = r \quad (\#)$$

We introduce an auxiliary function for the proof:

For $\vec{e}_{-i} = (e_1', \dots, e_{i-1}', e_{i+1}', \dots, e_m')$, $e_j' \in \mathcal{E}^k$, and

$\vec{e}_{-i}' = (e_1'', \dots, e_{i-1}'', e_{i+1}'', \dots, e_m'')$, $e_j'' \in \mathcal{E}^{k-1}$, we define

$$\mathbb{I}(\vec{e}_{-i}, \vec{e}_{-i}') = \begin{cases} 1 & e_j'' = \mathbf{R}[e_j'] \text{ f.a. } j \neq i; \\ 0 & \text{otherwise.} \end{cases}$$

Since the regression is unique, given $\vec{e}_{-i}'$, there's only one tuple $\vec{e}_{-i}''$ with $\mathbb{I}(\vec{e}_{-i}, \vec{e}_{-i}'') = 1$. Thus when $\vec{e}_{-i}'$ is fixed,

$$\sum_{\vec{e}_{-i}''} \mathbb{I}(\vec{e}_{-i}, \vec{e}_{-i}'') = 1 \quad (*)$$

Back to (#), let $d' = \mathbf{R}[d]$, we have

$$\begin{aligned} r &= \sum_{(w, \vec{e}_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w, \vec{e}_{-i}) \cdot 1 \\ &= \sum_{(w, \vec{e}_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} (d(w, \vec{e}_{-i}) \cdot \sum_{\vec{e}_{-i}'} \mathbb{I}(\vec{e}_{-i}, \vec{e}_{-i}')) \quad (\text{Eq. } *) \\ &= \sum_{\{(w, \vec{e}_{-i}, \vec{e}_{-i}') \mid (w, \vec{e}_{-i}) \in \mathcal{W}_\alpha^{i, e_i}\}} d(w, \vec{e}_{-i}) \cdot \mathbb{I}(\vec{e}_{-i}, \vec{e}_{-i}') \\ &= \sum_{\{(w, \vec{e}_{-i}, \vec{e}_{-i}') \mid (w, \vec{e}_{-i}) \in \mathcal{W}_\alpha^{i, e_i}, e_j'' = \mathbf{R}[e_j'] \text{ f.a. } j \neq i\}} d(w, \vec{e}_{-i}) \quad (\text{Def. } \mathbb{I}) \end{aligned}$$

³By definition, the truth of $B_i(\alpha : r)$ is irrelevant to w and epistemic states of other agents except i . Thus we write $e_i \models B_i(\alpha : r)$ instead of $w, \vec{e} \models B_i(\alpha : r)$.

For $(w, \vec{e}_{-i}^*) \in \mathcal{W}_{\alpha}^{i, e_i}$, $e_i' = \mathbf{R}[e_i]$ and $e_i'' = \mathbf{R}[e_i^*]$ f.a. $j \neq i$, by induction hypothesis we have $w, e_1'', \dots, e_i'', \dots, e_m'' \models \alpha$, i.e. $(w, \vec{e}_{-i}'') \in \mathcal{W}_{\alpha}^{i, e_i'}$. Therefore,

$$\begin{aligned} r &= \sum_{\{(w, \vec{e}_{-i}^*) \mid (w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha}^{i, e_i'}, e_i'' = \mathbf{R}[e_i^*] \text{ f.a. } j \neq i\}} d(w, \vec{e}_{-i}^*) \quad (\text{IH}) \\ &= \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha}^{i, e_i'}} \left(\sum_{\{\vec{e}_{-i}^* \mid e_i'' = \mathbf{R}[e_i^*] \text{ f.a. } j \neq i\}} d(w, \vec{e}_{-i}^*) \right) \\ &= \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha}^{i, e_i'}} d'(w, \vec{e}_{-i}') \quad (\text{Def. 5}) \\ &\quad \text{Thus } \text{NORM}(d', \mathcal{W}_{\alpha}^{i, e_i'}, \mathcal{W}_{\text{TRUE}}^{i, e_i'} r) \text{ f.a. } d' \in e_i', \text{ i.e. } e_i' \models \mathbf{B}_i(\alpha : r) \end{aligned}$$

□

Now we can complete the semantics:

- For $e_i \in \mathcal{E}^{k'}$ s.t. $k' > k$, $w, \vec{e} \models \mathbf{O}_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ iff $w, e_1, \dots, \mathbf{R}[e_i], \dots, e_m \models \mathbf{O}_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$.

The result of Lem.1 can be extended to any wffs, including those with only-believing:

THEOREM 2. *Given $\vec{e}, \vec{e}'$ s.t. $e_i' = \mathbf{R}[e_i]$. For any formula α compatible with both $\vec{e}, \vec{e}'$, then $w, \vec{e} \models \alpha$ iff $w, \vec{e}' \models \alpha$*

The proof is similar to Lem 1. The only difference is the induction for $\mathbf{O}$ -formulae, which directly follows the definition of the semantics.

For a sentence α and a set of sentences Σ , we write $\Sigma \models \alpha$ (read as Σ logically entails α) to mean that for every model $(w, e_1, \dots, e_m)$ compatible with α and all sentences in Σ , if $w, e_1, \dots, e_m \models \alpha'$ f.a. $\alpha' \in \Sigma$, then $w, e_1, \dots, e_m \models \alpha$. We say α is valid (written as $\models \alpha$) if $\{\} \models \alpha$. When α is objective, we write $w \models \alpha$ instead of $w, e_1, \dots, e_m \models \alpha$. When α is subjective, we write $e_1, \dots, e_m \models \alpha$ (or $\vec{e} \models \alpha$). For i -subjective α , we write $e_i \models \alpha$.

3 PROPERTIES OF THE LOGIC

In this section, we study the properties of modalities $\mathbf{B}_i$ and $\mathbf{K}_i$. For $\mathbf{K}_i$, where the probabilistic belief is reduced to a categorical one, $\mathcal{OBL}_m$ satisfies the $KD45_n$ properties. The Barcan formulae[16] are also valid. For $\mathbf{B}_i$, we show that the degree of belief follows the laws of probability, the properties of introspection are also extended to more general cases. We provide proofs for some of the properties. The rest can be proved similarly.

3.1 Knowledge

$\mathcal{OBL}_m$ satisfies the $KD45_n$ properties:

- (Necessitation) If $\models \alpha$, then $\models \mathbf{K}_i \alpha$.
- (Consistency) $\models \mathbf{K}_i \alpha \supset \neg \mathbf{K}_i \neg \alpha$

PROOF. Suppose that $e_i \models \mathbf{K}_i \alpha$, then for any $d \in e_i$,

$$\begin{aligned} \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\neg \alpha}^{i, e_i}} d(w, \vec{e}_{-i}') &= \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\text{TRUE}}^{i, e_i} \setminus \mathcal{W}_{\alpha}^{i, e_i}} d(w, \vec{e}_{-i}') \\ &= 1 - \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha}^{i, e_i}} d(w, \vec{e}_{-i}') = 0 \end{aligned}$$

Thus $e_i \models \neg \mathbf{K}_i \neg \alpha$. □

- (Distribution) $\models \mathbf{K}_i \alpha \wedge \mathbf{K}_i (\alpha \supset \beta) \supset \mathbf{K}_i \beta$

PROOF. For any e_i compatible with $\mathbf{K}_i \alpha$ and $\mathbf{K}_i (\alpha \supset \beta)$, if $e_i \models \mathbf{K}_i \alpha \wedge \mathbf{K}_i (\alpha \supset \beta)$, then for $d \in e_i$,

$$\begin{aligned} \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\beta}^{i, e_i}} d(w, \vec{e}_{-i}') &= \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha}^{i, e_i}} d(w, \vec{e}_{-i}') \\ &\quad + \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha \supset \beta}^{i, e_i}} d(w, \vec{e}_{-i}') \\ &\quad - \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha \wedge \beta}^{i, e_i}} d(w, \vec{e}_{-i}') \\ &= 1 + 1 - 1 = 1 \end{aligned}$$

i.e. $e_i \models \mathbf{K}_i \beta$. □

- (Pos. Introspection) $\models \mathbf{K}_i \alpha \supset \mathbf{K}_i \mathbf{K}_i \alpha$
- (Neg. Introspection) $\models \neg \mathbf{K}_i \alpha \supset \mathbf{K}_i \neg \mathbf{K}_i \alpha$

PROOF. Suppose that $e_i \models \neg \mathbf{K}_i \alpha$, then f.a. w and $\vec{e}_{-i}'$, $w, e_1', \dots, e_{i-1}', e_i, e_{i+1}', \dots, e_m' \models \neg \mathbf{K}_i \alpha$. Thus f.a. $d \in e_i$,

$$\sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\neg \mathbf{K}_i \alpha}^{i, e_i}} d(w, \vec{e}_{-i}') = \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\neg \mathbf{K}_i \alpha}^{i, e_i}} d(w, \vec{e}_{-i}') = 1$$

i.e. $e_i \models \mathbf{K}_i \neg \mathbf{K}_i \alpha$. □

Barcan formulae (both universal and existential versions):

- $\models \forall x. \mathbf{K}_i \alpha \supset \mathbf{K}_i \forall x. \alpha$

PROOF. Suppose that $e_i \models \forall x. \mathbf{K}_i \alpha$. By definition we have $e_i \models \mathbf{K}_i \alpha_n^x$ for any $n \in \mathcal{N}$. Then f.a. $d \in e_i$,

$$\sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\alpha_n^x}^{i, e_i}} d(w, \vec{e}_{-i}') = 1 \text{ f.a. } n \in \mathcal{N}$$

Namely,

$$\sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\neg \alpha_n^x}^{i, e_i}} d(w, \vec{e}_{-i}') = 0 \text{ f.a. } n \in \mathcal{N}$$

Therefore

$$\begin{aligned} \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\forall x. \alpha}^{i, e_i}} d(w, \vec{e}_{-i}') &= 1 - \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\exists x. \neg \alpha}^{i, e_i}} d(w, \vec{e}_{-i}') \\ &\geq 1 - \sum_{n \in \mathcal{N}} \sum_{(w, \vec{e}_{-i}') \in \mathcal{W}_{\neg \alpha_n^x}^{i, e_i}} d(w, \vec{e}_{-i}') = 1 \end{aligned}$$

Thus $e_i \models \mathbf{K}_i \forall x. \alpha$. □

It is worth mentioning that the converse result also holds, i.e. $\models \mathbf{K}_i \forall x. \alpha \supset \forall x. \mathbf{K}_i \alpha$.

- $\models \exists x. \mathbf{K}_i \alpha \supset \mathbf{K}_i \exists x. \alpha$

The proof is of the same spirit as the universal one.

- $\not\models \mathbf{K}_i \exists x. \alpha \supset \exists x. \mathbf{K}_i \alpha$.

Let $w, w' \in \mathcal{W}$ satisfy $P(n_1) \in w, P(n') \notin w$ f.a. $n' \neq n_1$, $P(n_2) \in w', P(n') \notin w'$ f.a. $n' \neq n_2$. Let $d_i \in \mathcal{D}^1$ assign weight 0.5 to both w and w' . Other worlds are assigned 0. Then $\{d_i\}$ satisfies $\mathbf{K}_i \exists x. \alpha$ but not $\exists x. \mathbf{K}_i \alpha$.

3.2 Degree of Belief

For any $i \in Ag$ and formula α, β ,

- if α is valid, then $\models B_i(\alpha: 1)$
- $\models B_i(\alpha: r) \supset \neg B_i(\alpha: r')$ for $r' \neq r$
- if $\models \alpha \equiv \beta$, then $\models B_i(\alpha: r) \equiv B_i(\beta: r)$ for any r .
- $\models B_i(\alpha: r) \supset B_i(\neg\alpha: 1 - r)$
- $\models B_i(\alpha \wedge \beta: r) \wedge B_i(\alpha \wedge \neg\beta: r') \supset B_i(\alpha: r + r')$
- $\models B_i(\alpha: r) \wedge B_i(\beta: r') \wedge B_i(\alpha \wedge \beta: r'') \supset B_i(\alpha \vee \beta: n)$
where n is a standard name of sort number and $n = r + r' - r''$.

Most of the properties can be proved in a way similar to those of the previous subsection. Here we only provide the proof of the last one:

PROOF. Suppose that $e_i \models B_i(\alpha: r) \wedge B_i(\beta: r') \wedge B_i(\alpha \wedge \beta: r'')$, then f.a. $d \in e_i$, it holds $\sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w, \vec{e}'_{-i}) = r$, $\sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\beta^{i, e_i}} d(w, \vec{e}'_{-i}) = r'$, $\sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_{\alpha \wedge \beta}^{i, e_i}} d(w, \vec{e}'_{-i}) = r''$. By applying the laws of set,

$$\mathcal{W}_{\alpha \vee \beta}^{i, e_i} = \mathcal{W}_\alpha^{i, e_i} + \mathcal{W}_{\neg\alpha \wedge \beta}^{i, e_i} = \mathcal{W}_\alpha^{i, e_i} + \mathcal{W}_\beta^{i, e_i} - \mathcal{W}_{\alpha \wedge \beta}^{i, e_i}$$

Thus $\sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_{\alpha \vee \beta}^{i, e_i}} d(w, \vec{e}'_{-i}) = r + r' - r''$ $\square$

Results of introspection also hold for degrees of belief. Furthermore, it can be extended to arbitrary i -subjective sentences.

- $\models B_i(\alpha: r) \supset K_i B_i(\alpha: r)$
- $\models \neg B_i(\alpha: r) \supset K_i \neg B_i(\alpha: r)$
- For any i -subjective formula α , $\models \alpha \supset K_i \alpha$

PROOF. Suppose that α is i -subjective and $e_i \models \alpha$. Since α is i -subjective, given a model $(w, e_1, \dots, e_m)$, the truth value is irrelevant to w or epistemic states other than e_i . For any $d \in e_i$,

$$\sum_{\{w, \vec{e}'_{-i} \mid w, \vec{e}'_{-i} \models \alpha\}} d(w, \vec{e}'_{-i}) = \sum_{\{w, \vec{e}'_{-i}\}} d(w, \vec{e}'_{-i}) = 1$$

Thus $e_i \models K_i \alpha$ $\square$

4 ONLY-BELIEVING

We discuss the properties of only-believing in this section. First, we examine the relation between only-believing up to different depths and demonstrate how a hierarchy of only-believing is built. For cases of only-believing when the argument is a group of i -objective formulae, we show the uniqueness of the model and the nice properties it brings. For sentences beyond i -objective, we demonstrate how certain types of autoepistemic reasoning can be modelled, and how the specification of only-believing contributes to the expressiveness.

Intuitively, what being only-believed should be believed at first:

PROPOSITION 1. $\models O_i^{(k)}(\alpha_1: r_1, \dots, \alpha_l: r_l) \supset \bigwedge_{j=1}^l B_i(\alpha_j: r_j)$

If an agent only believes α with degree r up to a certain depth, then she should also only believe it up to a lower (but compatible) depth. The converse result does not necessarily hold. Formally,

PROPOSITION 2. $\models O_i^{(k+1)}(\alpha: r) \supset O_i^{(k)}(\alpha: r)$.

PROOF. The proof is similar to Lem. 1 with function $\mathbb{I}$ as used in the lemma. Suppose that $e_i \models O_i^{(k+1)}(\alpha: r)$, w.l.o.g. we assume that $e_i \in \mathcal{E}^{(k+1)}$. By Thm 2 it suffices to prove that $e'_i \models O_i^{(k)}(\alpha: r)$ for $e'_i = \mathbf{R}[e_i]$. For $d \in \mathcal{D}^{(k+1)}$, let d' be the k -distribution s.t. $d' = \mathbf{R}[d]$. By the semantics, $d \in e_i$ iff

$$\begin{aligned} & \sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w, \vec{e}'_{-i}) = r \\ & \sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w, \vec{e}'_{-i}) \cdot 1 \\ &= \sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} (d(w, \vec{e}'_{-i}) \cdot \sum_{\vec{e}''_{-i}} \mathbb{I}(\vec{e}'_{-i}, \vec{e}''_{-i})) \quad (\text{Eq. } *) \\ &= \sum_{\{(w, \vec{e}'_{-i}, \vec{e}''_{-i}) \mid (w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}, e'_j = \mathbf{R}[e'_j] \text{ f.a. } j \neq i\}} d(w, \vec{e}'_{-i}) \\ &= \sum_{\{(w, \vec{e}'_{-i}, \vec{e}''_{-i}) \mid (w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}, e'_j = \mathbf{R}[e'_j] \text{ f.a. } j \neq i\}} d(w, \vec{e}'_{-i}) \quad (\text{Thm. 2}) \\ &= \sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e'_i}} \left(\sum_{\{\vec{e}''_{-i} \mid e'_j = \mathbf{R}[e'_j] \text{ f.a. } j \neq i\}} d(w, \vec{e}'_{-i}) \right) \\ &= \sum_{(w, \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e'_i}} d'(w, \vec{e}'_{-i}) = r \quad (\text{Def. 5}) \end{aligned}$$

By Def. 5, $d' \in e'_i$ iff $d \in e_i$. Thus $e'_i \models O_i^{(k)}(\alpha: r)$. $\square$

There exists α s.t. $\not\models O_i^{(k)}(\alpha: r) \supset O_i^{(k+1)}(\alpha: r)$. An example can be easily constructed: Let $e_i \in \mathcal{E}^2$ and $e_i \models O_1^{(2)}(p \wedge K_2 p)$, then $e_i \models O_1^{(1)} p$ but $e_i \not\models O_1^{(2)} p$.

The proposition demonstrates that from $O_i^{(k)}$ to $O_i^{(k+1)}$ it is indeed a more precise specification of an agent's only-believing.

There are unsatisfiable O -formulae, for example $O_i^{(k)}(p \wedge K_i \neg p)$. Fortunately, for a large fragment of the language, an O -formula is satisfiable when the corresponding belief formulae are satisfiable:

PROPOSITION 3. Let $\alpha_1, \dots, \alpha_l$ be i -objective. If $\bigwedge_{j=1}^l B_i(\alpha_j: r_j)$ is satisfiable, then $O_i^{(k)}(\alpha_1: r_1, \dots, \alpha_l: r_l)$ is satisfiable.

4.1 Unique Model and Properties

In general, an $O_i^{(k)}$ -formula could be satisfied by more than one epistemic state. For example $O_1^{(1)}((p \wedge K_1 p) \vee (\neg p \wedge K_1 \neg p))$. For i -objective sentence α , however, there is a unique k -epistemic state $e_i \in \mathcal{E}^k$ which satisfies $O_i^{(k)} \alpha$:

LEMMA 2. Given i -objective α and k s.t. $K_i \alpha$ is satisfiable and $O_i^{(k)} \alpha$ is a wff, there is precisely one $e_i \in \mathcal{E}^k$ s.t. $e_i \models O_i^{(k)} \alpha$.

PROOF. Prop. 3 proves the existence of e_i . Given $e_i \in \mathcal{E}^k$, by the definition $e_i \models O_i^{(k)} \alpha$ iff f.a. $d \in \mathcal{D}^k$, $d \in e_i$ iff

$$\sum_{(w', \vec{e}'_{-i}) \in \mathcal{W}_\alpha^{i, e_i}} d(w', \vec{e}'_{-i}) = 1$$

Since α is i -objective, if $w', e'_1, \dots, e_i, \dots, e'_m \models \alpha$, then f.a. $\tilde{e}_i$, $w', e'_1, \dots, \tilde{e}_i, \dots, e'_m \models \alpha$. For any $\tilde{e}_i \in \mathcal{E}^k$ s.t. $\tilde{e}_i \models O_i^{(k)}\alpha$, it is trivial that f.a. $d \in \mathcal{D}^k$, $d \in e_i$ iff $d \in \tilde{e}_i$, i.e. $e_i = \tilde{e}_i$. $\square$

For a model of a depth greater than needed, the uniqueness of the model can be interpreted as follows: Suppose that $e_i \in \mathcal{E}^k$ is the unique model satisfying $O_i^{(k)}\alpha$. Then for all $e_i^* \in \mathcal{E}^{k'}$ s.t. $k' > k$ and $e_i^* \models O_i^{(k)}\alpha$, e_i^* will reduce to e_i after finite steps of regression:

LEMMA 3. *Given i -objective α and k s.t. $K_i\alpha$ is satisfiable and $O_i^{(k)}\alpha$ is a wff. Let $e_i \in \mathcal{E}^k$ be the model s.t. $e_i \models O_i^{(k)}\alpha$. For any e_i^* such that $e_i^* \models O_i^{(k)}\alpha$, $e_i = \mathbf{R}^{(l)}[e_i^*]$ for some l .*

PROOF. For any $k' > k$ and $e_i^* \in \mathcal{E}^{k'}$, e_i^* reduces to $\tilde{e}_i \in \mathcal{E}^k$ after $k' - k$ steps of regression. Given $e_i^* \models O_i^{(k)}\alpha$, by applying Thm. 2 we have $\tilde{e}_i \models O_i^{(k)}\alpha$. By Lem. 2, $\tilde{e}_i$ is unique, i.e. $\tilde{e}_i = e_i$. $\square$

In general, the results can be extended to the case with multiple beliefs:

THEOREM 3. *Given i -objective $\alpha_1 \dots \alpha_l$ s.t. $\bigwedge_j B_i(\alpha_j : r_j)$ is satisfiable and $O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ is a wff, there is precisely one $e_i \in \mathcal{E}^k$ such that $e_i \models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$. For any e_i^* such that $e_i^* \models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$, $e_i = \mathbf{R}^{(l)}[e_i^*]$ for some l .*

The proof is similar to Lem.2 and Lem.3.

A nice property of the uniqueness is that, given what is only-believed by an agent, everything else being believed or not believed by the agent is logically implied. Formally, we have the following results:

THEOREM 4. *Given i -objective formulae $\alpha_1, \dots, \alpha_l$ and arbitrary β s.t. $O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ is satisfiable, $\text{dep}[B_i(\beta : r'), i] \leq k$. Then for any $0 < r' < 1$, either $B_i(\beta : r')$ or $\neg B_i(\beta : r')$ is entailed by $O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$, and*

- $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset B_i(\beta : r')$ iff $\models \bigwedge_j B_i(\alpha_j : r_j) \supset B_i(\beta : r')$
- $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset \neg B_i(\beta : r')$ iff $\not\models \bigwedge_j B_i(\alpha_j : r_j) \supset B_i(\beta : r')$

PROOF. By Thm.3, it exists a unique $e_i \in \mathcal{E}^k$ such that $e_i \models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$. If $\bigwedge_j B_i(\alpha_j : r_j) \supset B_i(\beta : r')$ is valid, by Prop.1, $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset B_i(\beta : r')$. When $\bigwedge_j B_i(\alpha_j : r_j) \supset B_i(\beta : r')$ is invalid, there exists $d_i \in \mathcal{D}^k$ s.t. $\{d_i\} \models \bigwedge_j B_i(\alpha_j : r_j) \wedge \neg B_i(\beta : r')$. By the semantics $d_i \in e_i$. Hence $e_i \models \neg B_i(\beta : r')$ and $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset \neg B_i(\beta : r')$. For any $e_i' \in \mathcal{E}^{k'}$ s.t. $k' > k$, $e_i' \models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$. By Thm. 2 and 3, $e_i' \models B_i(\beta : r')$ iff $e_i \models B_i(\beta : r')$. $\square$

THEOREM 5. *Given i -objective formulae $\alpha_1, \dots, \alpha_l$ and arbitrary β , s.t. $O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l)$ is satisfiable, $\text{dep}[K_i\beta, i] \leq k$, then either $O_i^{(k)}(\alpha : r) \supset K_i\beta$ or $O_i^{(k)}(\alpha : r) \supset \neg K_i\beta$ is valid, and*

- $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset K_i\beta$ iff $\models \bigwedge_j B_i(\alpha_j : r_j) \supset K_i\beta$ or $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset \beta$

- $\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset \neg K_i\beta$ iff $\not\models \bigwedge_j B_i(\alpha_j : r_j) \supset K_i\beta$ and $\not\models O_i^{(k)}(\alpha_1 : r_1, \dots, \alpha_l : r_l) \supset \beta$

Given distinct propositions p, q , to determine whether $O_i^{(k)}p$ entails $K_i\neg K_iq$, it suffices to check the validity of $K_i p \supset K_i\neg K_iq$ and $O_i^{(k)}p \supset \neg K_iq$. The latter is valid since $\not\models K_i p \supset K_iq$ and $\not\models O_i^{(k)}p \supset q$. Thus $\models O_i^{(k)}p \supset K_i\neg K_iq$.

Note that this is not true for K_i , e.g. neither $K_i p \supset K_i\neg K_iq$ nor $K_i p \supset \neg K_i\neg K_iq$ is valid. Also, we would like to reiterate that the results of Thm.4 and Thm.5 hold only under the restriction of depth and do not hold when $\text{dep}[K_i\beta, i] > k$. For example, both $O_1^{(1)}p \wedge K_1(p \wedge K_2p)$ and $O_1^{(1)}p \wedge K_1(p \wedge \neg K_2p)$ are satisfiable.

If β is also i -objective, we can reduce the use of the outermost modalities:

COROLLARY 1. *Given i -objective α, β , s.t. $O_i^{(k)}\alpha$ is satisfiable, then*

- $\models O_i^{(k)}\alpha \supset K_i\beta$ iff $\models \alpha \supset \beta$
- $\models O_i^{(k)}\alpha \supset \neg K_i\beta$ iff $\not\models \alpha \supset \beta$

4.2 Default Reasoning

Levesque and Lakemeyer have already shown how defaults can be encoded in terms of only-knowing [24]. We can go beyond that: Besides the well-known “birds can fly” default, our account can also express defaults about what another agent believes.

EXAMPLE 2. *Let predicate Fair(x) mean “(The coin) x is fair”. Let δ denote the sentence*

$$\forall x. [\text{Coin}(x) \wedge \neg \exists r. (r > 0 \wedge B_1(\neg K_2 \text{Fair}(x) : r)) \supset K_2 \text{Fair}(x)]$$

i.e. for any coin, agent 2 believes that the coin is fair unless it is believed to a positive degree that she doesn't believe it. Let KB be the sentence $\{\text{Coin}(\text{silver})\}$, then the following sentences are valid:

- $O_1^{(2)}(\text{KB} \wedge \delta) \supset K_1 K_2 \text{Fair}(\text{silver})$
- $O_1^{(2)}(\text{KB} \wedge \delta \wedge K_2 \text{Fair}(\text{silver})) \supset K_1 K_2 \text{Fair}(\text{silver})$
- $O_1^{(2)}(\text{KB} \wedge \delta \wedge \neg K_2 \text{Fair}(\text{silver})) \supset K_1 \neg K_2 \text{Fair}(\text{silver})$

We show the validity of the first sentence. For simplicity, only two agents are considered.

PROOF. Let $e_1 \models O_1^{(2)}(\text{KB} \wedge \delta)$, w.l.o.g. $e_1 \in \mathcal{E}^2$. We first prove $e_1 \models \neg \exists r. r > 0 \wedge B_1(\neg K_2 \text{Fair}(\text{silver}) : r)$ by contradiction. Assuming that the opposite holds, i.e. there is a number $n > 0$ s.t. $e_1 \models B_1(\neg K_2 \text{Fair}(\text{silver}) : n)$. By the definition of $O_1^{(2)}$, f.a. $d \in \mathcal{D}^2$, $d \in e_1$ iff $\text{NORM}(d, \mathcal{W}_{\text{KB} \wedge \delta}^{1, e_1}, \mathcal{W}_{\text{TRUE}}, 1)$. Namely, $d \in e_1$ iff f.a. $w \in \mathcal{W}$ and $e_2 \in \mathcal{E}^1$, if $w, e_1, e_2 \models \text{KB} \wedge \delta$ then $d(w, e_2) = 0$. We select two states $e_2', e_2'' \in \mathcal{E}^1$ s.t.

$$\begin{aligned} e_2' &\models \neg K_2 \text{Fair}(\text{silver}) \wedge \forall x. [x \neq \text{silver} \supset K_2 \text{Fair}(x)] \\ e_2'' &\models \forall x. K_2 \text{Fair}(x) \end{aligned}$$

The choice of e_2', e_2'' is arbitrary as long as the conditions are satisfied. Let w' be a world s.t. $w' \models \text{KB}$, n' a number in $[0, 1]$ and $n' \neq n$. We define a 2-distribution d_1 as follows:

$$d_1(w, e_2) = \begin{cases} n' & w = w', e_2 = e_2'; \\ 1 - n' & w = w', e_2 = e_2''; \\ 0 & \text{otherwise.} \end{cases}$$

With the assumption that $e_1 \models B_1(\neg K_2 \text{Fair}(\text{silver}) : n)$, we conclude that $w'e_1, e'_2 \models KB \wedge \delta$ and $w'e_1, e''_2 \models KB \wedge \delta$. Thus f.a. $w, e_2, d_1(w, e_2) = 0$ if $w, e_1, e_2 \models KB \wedge \delta$, i.e. $d_1 \in e_1$. Base on the semantics $e_1 \models \neg B_1(\neg K_2 \text{Fair}(\text{silver}) : n)$, which contradicts the assumption.

Since sentence $\neg \exists r. r > 0 \wedge B_1(\neg K_2 \text{Fair}(\text{silver}) : r)$ is 1-subjective, $e_1 \models K_1(\neg \exists r. r > 0 \wedge B_1(\neg K_2 \text{Fair}(\text{silver}) : r))$. With the distribution rule, we prove that $e_1 \models K_1 K_2 \text{Fair}(\text{silver})$.

We also check the existence of such a model. Let e_1 be precisely the set of all distributions d_1 such that $d_1(w, e_2) = 0$ f.a. (w, e_2) which satisfies at least one of the conditions:

- $w \models \neg \text{Coin}(\text{silver})$ or $e_2 \models \neg K_2 \text{Fair}(\text{silver})$;
- $w \models \text{Coin}(n)$ and $e_2 \models \neg K_2 \text{Fair}(n)$ for $n \neq \text{silver}$.

By the semantics, it can be proved that $e_1 \models O_1^{(2)}(KB \wedge \delta)$. $\square$

5 RELATION TO OTHER LOGICS

5.1 $OB\mathcal{L}$ is part of $OB\mathcal{L}_m$

$OB\mathcal{L}$ is a first-order, single-agent account of subjective probability and only-believing[6], which is an extension of the logic $O\mathcal{L}$ proposed by Levesque [23, 24]. For cases where at most one agent is involved, $OB\mathcal{L}_m$ reduces to $OB\mathcal{L}$ —with exceptions: In $OB\mathcal{L}$, the empty epistemic state is legal, and distributions, where the weightings of worlds do not form a proper probability distribution, are not excluded. The former setting breaks the consistency, e.g. $K_{\text{True}} \wedge K_{\text{False}}$ is satisfiable in $OB\mathcal{L}$, and the latter breaks the necessitation rules, e.g. True is valid in $OB\mathcal{L}$ but K_{True} is not. We argue that omitting these structures in the semantics has no loss of expressiveness or generality. On the other hand, by the exclusion of these structures, the logic will obtain the complete $KD45_n$ properties, while only simple, straightforward restrictions are required. Formally, we have the following result:

THEOREM 6. Let $\models_{OB\mathcal{L}}$ denote the satisfaction relation in $OB\mathcal{L}$, for any well-formed $OB\mathcal{L}_m$ sentence α where no modalities appear except B_i and $O_i^{(1)}$ for an agent i . Let α' be the sentence to replace every occurrence of B_i in α with B , and replace $O_i^{(1)}$ with O , then

$$\models \alpha \text{ iff } K_{\text{True}} \wedge \neg K_{\text{False}} \models_{OB\mathcal{L}} \alpha'$$

Examples include the properties discussed in Sections 3 and 4.

5.2 Relation to $ON\mathcal{L}_n$

$ON\mathcal{L}_n$ proposed by Belle and Lakemeyer[3] is a first-order modal language for multi-agent only-knowing, where operators L_i, N_i are used to describing what agent i “at least knows” and “at most knows”. Only-knowing is considered as an abbreviation, e.g. $O_i \alpha \equiv L_i \alpha \wedge N_i \neg \alpha$. The k -distributions or k -epistemic states in this paper are inspired by their work. One might notice that in $ON\mathcal{L}_n$, the superscript (k) is not required for only-knowing. In their account, only-knowing can not precisely capture beliefs and non-beliefs of sentences with greater depth. For instance, neither $O_1 p \supset K_1 K_2 q$ nor $O_1 p \supset \neg K_1 K_2 q$ is valid in $ON\mathcal{L}_n$. We resolve this problem by specifying the depth of only-knowing with the superscript. For $k \geq 2$, $O_1^{(k)} p$ means that p is all agent 1 knows up to depth k , where the depth for $K_1 K_2 q$ is also included. We have $\models O_1^{(k)} p \supset \neg K_1 K_2 q$.

6 OTHER RELATED WORK

In the introduction, we have discussed related work about only-knowing and default reasoning. We provide more about notions related to only-knowing as well as unifying logic and probability.

The notion of only-knowing did not emerge in isolation. The idea of involving accessible and inaccessible worlds is also adopted by Humberstone [17], Ben-David and Gafni [7]. Characteristics of only-knowing are shared by similar concepts such as *total knowledge* from Pratt-Hartmann [30] and *minimal knowledge* from Halpern and Moses [15].

Numerous pieces of literature can be found in reasoning about knowledge and probabilities. For example the work from Nilsson [28]. First-order accounts of probabilities are discussed by both Bacchus [2] and Halpern [12]. For multi-agent scenarios, Fagin and Halpern proposed a framework based on Kripke semantics to reason about higher-order probabilities [11]. Their scheme was later extended to express common beliefs [32]. For the interest of AI, there are approaches combining first-order logic and probabilistic graphical models, such as first-order Belief network [29], Markov Logic Network [10], etc.

7 CONCLUSION

The main results of this work are as follows: We proposed a first-order modal logic for multi-agent only-believing, which for the first time generalizes Levesque’s semantics to a multi-agent, probabilistic scenario. While $OB\mathcal{L}_m$ is downward compatible with $OB\mathcal{L}$ and $O\mathcal{L}$, the ability to capture the beliefs and non-beliefs of a knowledge base is lifted so that meta-beliefs involving many agents can also be considered. We also explored the capability of default reasoning. Compared with previous works, $OB\mathcal{L}_m$ is strictly more expressive.

Several aspects of future work are considered: We plan to augment the account to express common belief. Many of the previous works with similar designs on semantics support the possibility of such an extension. In particular, recent work from Cramer et. al.[9] shows that for a not fully-introspective account of categorical knowledge, combining common knowledge and only-knowing up to arbitrary depth can be achieved by at most $\omega^2 + 1$ layers of nesting in models.⁴ Since the language includes first-order logic as a fragment, reasoning in $OB\mathcal{L}_m$ is in general undecidable. A feasible solution is to restrict the types of knowledge to be represented, or reconsider the form of reasoning [22]. We also intend to consider multi-agent only-believing in a dynamic system. Belle and Lakemeyer proposed an account for categorical knowledge [4], yet it was restricted to actions with fully observable, deterministic effects. For a multi-agent scenario with incomplete information, action taken by an agent may not be observable for another. Uncertainty in action should be modelled in a probabilistic form.

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⁴ ω is a *limit ordinal* number greater than all natural numbers. The notion of ordinal numbers is a generalization of natural numbers in set theory.

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